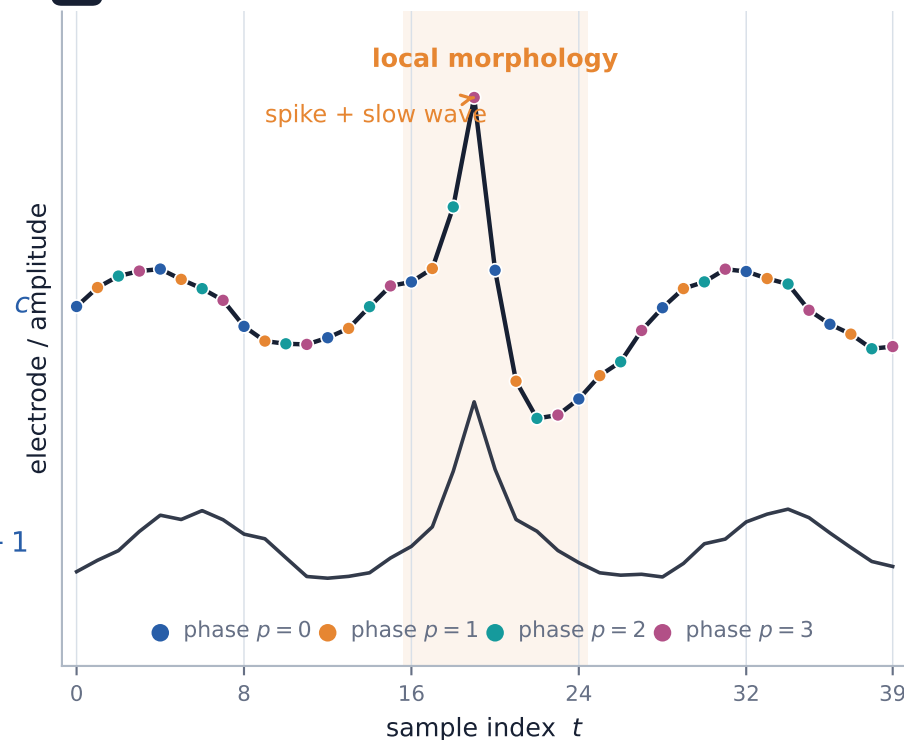


Why Phase-Folded EEG Fits the Receptive Fields of a 2D CNN

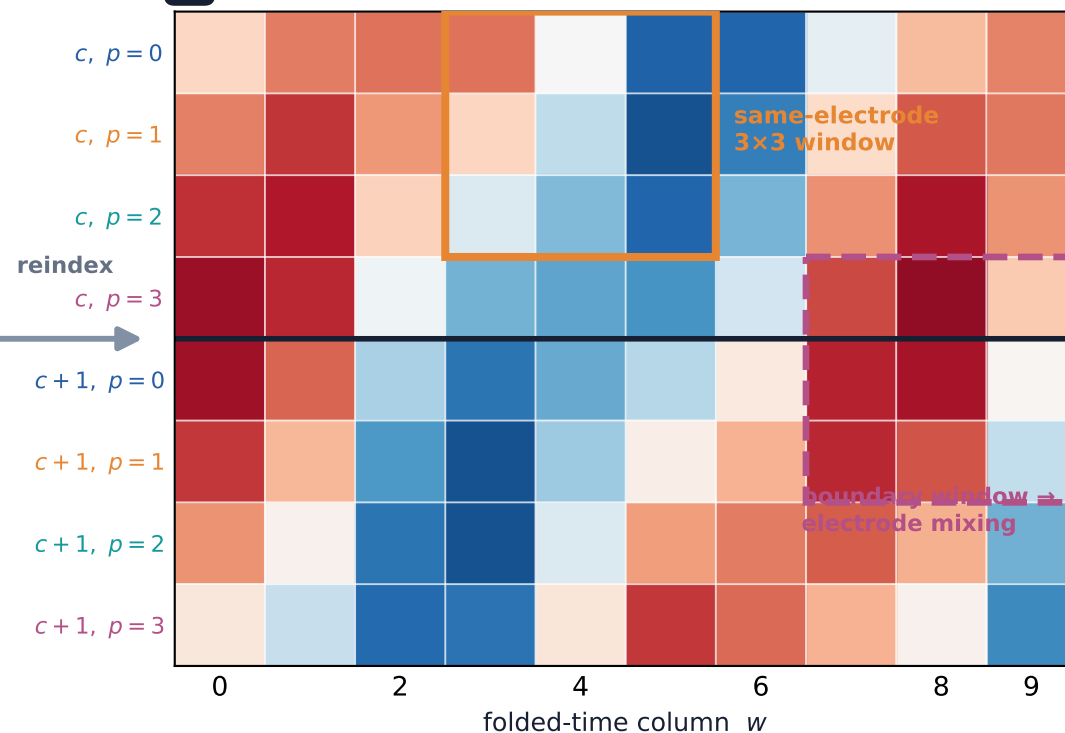
A lossless permutation exposes polyphase temporal structure as local 2D neighborhoods, then CNN depth integrates progressively longer and wider EEG context.

A Raw EEG is one-dimensional



Input geometry: $X \in \mathbb{R}^{C \times T}$

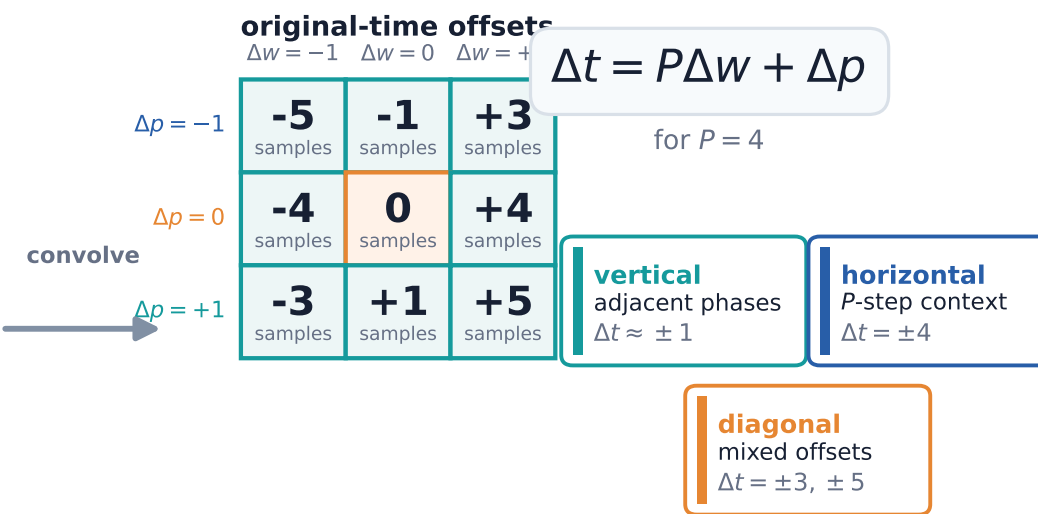
B Lossless phase folding ($P = 4$)



$$I[cP + p, w] = X[c, wP + p]$$

$[C, T] \rightarrow [CP, T/P]$ · permutation only

C A 2D kernel becomes multi-lag temporal filtering

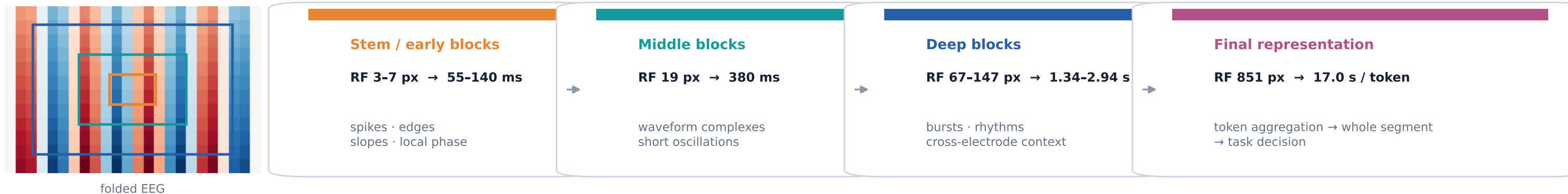


One local square sees slopes, phases, and dilated context at once.

D EfficientNet-B0 grows from waveform morphology to global context

Temporal span: $R_t(P) = P(R_w - 1) + \min(R_h, P)$ (within one electrode)

Example scales use $P = 4$ and 200 Hz; vertical RF growth also crosses electrode groups.



Core insight: folding changes adjacency—not information—so pretrained visual filters receive an EEG geometry they can exploit.